

SOLVING REAL-WORLD PROBLEMS INVOLVING NUMBERS EXPRESSED IN SCIENTIFIC NOTATION



LESSON INTRODUCTION

MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

LEARNING TARGETS

- I can solve real-world problems involving operations with numbers expressed in scientific notation.
- I can express answers in scientific notation using significant digits.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

WORKING TOWARD

MA.9.12.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational number exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

ESSENTIAL QUESTIONS

- What are significant digits?
- When is zero is a significant digit and when is it not?
- How do you solve real-world problems involving operations with numbers expressed in scientific notation?

LESSON NARRATIVE

Students initially develop an understanding of how to express answers in scientific notation using significant digits. Next, they practice solving real-world problems involving operations with numbers expressed in scientific notation. Students see the utility of the learning target through a variety of real-world examples.

COMPONENT SUMMARY

3.8.1 WARM UP (~5 MIN.)

Solve puzzle involving operations with numbers expressed in scientific notation

3.8.3 GUIDED INSTRUCTION (10 MIN.)

Introduction to solving real-world problems involving operations with numbers expressed in scientific notation

3.8.5 WRAP-UP (~5 MIN.)

Reflect on mastery of unit learning targets

3.8.2 GUIDED INSTRUCTION (15-20 MIN.)

Introduction to adding and subtracting numbers written in scientific notation and how to determine significant digits

3.8.4 YOUR TURN! (10 MIN.)

Practice solving real-world problems involving operations with numbers expressed in scientific notation

RESOURCES

GLOSSARY

Key Terms:

- Significant Digits

Additional Terms:

- N/A

MATERIALS

- (Optional) Chart Paper & Markers

ADDITIONAL PREPARATION

- Consider adding significant digits to a classroom Word Wall and creating and displaying Anchor Charts with each operation.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think zeros are never significant digits.
- Students may not understand why they cannot add or subtract the mantissa when the powers of ten are different.
- Students may not understand why they should apply the product of powers law and add the powers of 10.
- Students may not understand why they should apply the quotient of powers law and subtract the powers of 10.
- Students may not understand how to rewrite their answers in scientific notation.

THINGS TO CONSIDER

- In this lesson, students will perform all four operations with numbers written in scientific notation. Consider reviewing each operation with an example.
- If time is an issue, consider assigning all or part of 3.8.4 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

In 3.8.2 Guided Instruction students have the opportunity to compare and connect their strategies and answers. These discussions will help students build confidence and understanding with the lesson learning targets.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

The lesson learning targets are embedded in real-world contexts in 3.8.2 Guided Instruction, 3.8.3 Guided Instruction, and 3.8.4 Your Turn! These connections help students see the utility of performing operations on numbers expressed in scientific notation and expressing answers in scientific notation using significant digits.

DIFFERENTIATE INSTRUCTION

Provide a visual entry point in 3.8.2 Guided Instruction by prompting students to circle the significant digits and write a check in the columns that state which rules they used when making their decisions. Partner discussions about the significant digits provide an auditory entry point.

	Non-zero digits always significant	Zeros between non-zero digits always significant	Zeros to left of non-zero digits not significant	All zeroes to the right are significant if there is a decimal point
0.0807	x	x		
13.401	x	x		
1.990	x		x	
5.5	x			
367.0	x			x

3.8.1 WARM-UP: CRITICAL THINKING

TEACHER MOVES

Consider having a variety of students share how they obtained their solutions. After each student shares ask the class if they agree or disagree and why. Ask students whether it is better to put the larger/smaller digits in the mantissa or as the power of 10 for each operation and why.



3.8.1 Warm-Up: Critical Thinking

- Using the digits 1 – 9 at most twice each, create make the sum of the four expressions that has the greatest possible value.

6×10^7	+	6×10^7	=	1.2×10^8
4×10^5	–	2×10^1	=	3.9998×10^5
8×10^9	×	8×10^9	=	6.4×10^{19}
4×10^5	÷	2×10^1	=	2.0×10^4
Total				6.4×10^{19}



3.8.2 Guided Instruction: Understanding Significant Digits in Scientific Notation

1. The table shows the distance from Earth, in kilometers (km), to the nearest two planets.

Planet	Distance (km)	Scientific Notation
Venus	38 million	3.8×10^7
Mars	55.7 million	5.57×10^7

Jackson was asked to find the difference between the two distances by rounding to the nearest whole number to make the values easier to work with.

$$(6 \times 10^7) - (4 \times 10^7) = (2 \times 10^7) \text{ km}$$

Jocelyn was asked to perform the calculation without rounding.

$$(5.57 \times 10^7) - (3.8 \times 10^7) = (1.77 \times 10^7) \text{ km}$$

- a. What is the difference, in km, between the two answers?
 2.3×10^6
- b. Discuss with your partner why Jocelyn's answer is more precise.
Because Jocelyn did not round the original distances, her answer is more precise.



Significant digits are non-zero digits of a number and the zeros that are included between them or any trailing zeros that are considered to be precise.

- c. Based on the definition, how many significant digits are in the representations of the distances for Mars and Venus?
Venus has 2, and Mars has 3.

3.8.2 GUIDED INSTRUCTION: UNDERSTANDING SIGNIFICANT DIGITS IN SCIENTIFIC NOTATION

DIFFERENTIATE INSTRUCTION

Partner discussions about the significant digits provide an auditory entry point.

ENRICHMENT

Is the number of significant digits the same for the standard form and scientific notation of a number? How can you remember how to determine which digits are significant?

Examples

2027 has 4 significant digits.	00052000 has 2 significant figures.
00.00800 has 3 significant digits.	00035000.0 has 6 significant figures.

There are three rules on determining how many significant figures are in a number.

- Non-zero digits are always significant.
- Any zeros between two significant digits are significant.
- A final zero or trailing zeros in the decimal portion ONLY are significant.

Sometimes answers written in scientific notation are written using a specific number of significant digits. If two numbers are written with different precision, then the precision is the least number of significant digits.

- d. Discuss with your partner the given values of the distance from Earth to Mars and Venus. Determine how many significant digits should be used to represent the difference between Mars and Venus.
There are 2 significant digits in the difference as seen in the answer in scientific notation.

3.8.2 GUIDED INSTRUCTION: UNDERSTANDING SIGNIFICANT DIGITS IN SCIENTIFIC NOTATION (CONTINUED)

DIFFERENTIATE INSTRUCTION

This problem provides a visual entry point as students circle the significant digits and write a check in the columns that state which rules they used when making their decisions.

2. Circle the significant digits in each number. Place a check in the column(s) to reflect which rule(s) you used when making your decision.

	Non-Zero Digits Always Significant	Zeros Between Non-Zero Digits Always Significant	Zeros to the Left of Non-Zero Digits Not Significant	All Zeros to the Right are Significant if There is a Decimal Point
0.0807	x	x	x	
13,401	x	x		
1,990	x			
5.5	x			
367.0	x			x

3. In 1966, the Apollo Guidance Computer was developed to make the calculations that would one day put humans on the Moon. In 2020, one phone company released the 12th smartphone in its latest series of smartphones.

Information about the memory chips in the devices is summarized in the table.

Device	Storage	Processing	Memory
Apollo 11 Guidance Computer	32 KB	1.042 MHz	2 KB
Smartphone	128 GB	2.7 MHz	6 GB

*For reference, storage and memory are measured in bytes (B), and processor speed is measured in hertz (Hz)

Kilobytes (KB) = 1,000 bytes (B)
Megahertz (MHz) = 1,000,000 hertz (Hz)
Gigabytes (G) = 1,000,000,000 bytes (B)
Terabytes (T) = 1,000,000,000,000 bytes (B)

TEACHER MOVES

Compare and Connect

Provide students the opportunity to compare and connect their strategies and answers. These discussions will help students build confidence and understanding with the lesson learning targets.

- Determine the combined storage and memory, in bytes, of the smartphone. Write your answer using scientific notation.
 1.34×10^{11}
- How much combined storage and memory, in bytes, are in 8 smartphones? Write your answer using scientific notation.
 1.072×10^{12}
- Determine how much faster, in hertz, the smartphone's processing speed is compared to the Apollo 11 Guidance Computer's processing speed. Write your answer using scientific notation.
 1.658×10^6

**3.8.3 Guided Instruction: Using Scientific Notation to Solve Real-World Problems**

1. The National Debt in 2020 was \$26.9 trillion. The population for the state of Florida in 2020 was estimated to be 21.9 million.
 - a. If the people living in Florida were to pay off the national debt, how much would each person pay?
 1.23×10^6
 - b. The population of North Bay in 2020 was 197,974. If the people living in North Bay were to pay off the national debt, how much would each person pay?
 1.36×10^8
 - c. It has been estimated that the national debt increases at a rate of about \$3.9 million each day. At this rate, how much will the debt increase in a year?
 1.4×10^9

3.8.3 GUIDED INSTRUCTION: USING SCIENTIFIC NOTATION TO SOLVE REAL-WORLD PROBLEMS

TEACHER MOVES

The lesson learning targets are embedded in real-world contexts. These connections help students see the utility of performing operations on numbers expressed in scientific notation and expressing answers in scientific notation using significant digits. Help students understand the context by asking them if they have ever borrowed money. Explain that national debt is like the country borrowing money.

3.8.4 YOUR TURN!

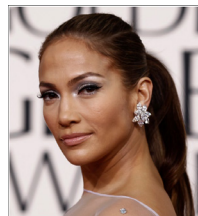
QUESTIONING

Challenge students to look up the earnings of their favorite celebrities. Then prompt them to write and solve problems involving the numbers expressed in scientific notation.



3.8.4 Your Turn!

- Jennifer Lopez changed the Hollywood landscape, single-handedly increasing Latino representation in film, television, and fashion. J.Lo, as she is known, has become one of the highest-paid Latina celebrities in Hollywood. She has multiple income streams.



The table summarizes her income from some of her recent jobs.

Job	Income
Judge and executive producer on the show "World of Dance" for 4 seasons	1.2×10^7 dollars per season
Jennifer Lopez clothing line, cosmetics, and fragrances	3×10^8 dollars in revenue
2012 "Dance Again" World Tour	5.1×10^7 dollars
2016-2018 "All I Have" concert residency in Las Vegas, Nevada	3.5×10^5 dollars per season

- What is the difference in Jennifer Lopez's income for "Dance Again" World Tour and 2016-2018 concert residency in Las Vegas, Nevada?

$$5.1 \times 10^7 - 3(3.5 \times 10^5) = 5.1 \times 10^7 - (1.05 \times 10^6)$$

$$\approx 5.0 \times 10^7$$
- What is sum of Jennifer Lopez's income from "World of Dance" and her clothing line, cosmetics, and fragrances?

$$4(1.2 \times 10^7) + 3.0 \times 10^8 = 3.48 \times 10^8$$

$$\approx 3 \times 10^8$$

DIFFERENTIATE INSTRUCTION

Help students determine which operation to perform by displaying a graphic organizer of key words from the problems that indicate which operation to use. For example:

!!!

Add: sum, total

Subtract: difference

Multiply: product, times Divide: quotient, how many times greater,

- How many times greater is Jennifer Lopez's income from her clothing line, cosmetics and fragrances in comparison to the income she made on her 2012 world tour?

$$\frac{3.0 \times 10^8}{5.1 \times 10^7} = 5.882352941$$

6 times greater

- Determine if Jennifer Lopez's income from 2016-2018 concert residency in Las Vegas, 2012 "Dance Again" World Tour, and show "World of Dance" is larger than her income from her clothing line, cosmetics, and fragrances.

$$4.8 \times 10^7 + 5.1 \times 10^7 + 1.05 \times 10^6 = 1.0005 \times 10^8$$

The income from her clothing line, cosmetics, and fragrances is greater than the other three.

- There are about (2.0×10^{11}) visible stars in a galaxy. There are about (8.0×10^{10}) galaxies in the universe. How many visible stars are there in the universe? Write your answer in scientific notation.

$$1.6 \times 10^{22}$$



3.8.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help someone.

Answers may vary.

Writing large numbers in scientific notation			
Writing small numbers in scientific notation			
Adding and subtracting numbers expressed in scientific notation			
Multiplying numbers expressed in scientific notation.			
Dividing numbers expressed in scientific notation.			
Solving real-world problems involving operations with numbers expressed in scientific notation.			

3.8.5 WRAP-UP: REFLECTION

QUESTIONING

If students finish early, prompt them to write and solve a problem to demonstrate their mastery of each learning target.